Tree Test Report

Carley Andelin and Ashley Thornton

Brigham Young University

CS 356: Advanced Techniques in HCI

Dr. Mike Jones and Professor Bob Carroll

October 12, 2026

1. Who Was in the Study

Sort	Participant	Interests	Selection Criteria
S1	Kate Thornton (P01)	Music, Foodie, Arts and Crafts, Social Experiences	Kate manages a team of employees and is looking for team building/party activity ideas.
S2	Spencer Tagge (P02)	Outdoorsman, Hiking, Sports, Concerts, Gaming	Spencer is curious and wants to “expand his experiences” to live life to the fullest.
S3	Isaiah Pack (P03)	Gaming, Cinema, weightlifting	Is a new father who is trying to reevaluate what family-friendly activities he should spend his free time doing
S4	Kyle Andelin (P04)	Programming, Current Events, Meeting new people	Kyle is a software developer that understands Information Architecture
S5	Savannah Sloan (P05)	Thrifting, doing nails, makeup, interior design	She is very trendy and picks up quick on things
S6	Sarah G (P06)	Religion, baking, hair care	Sarah has unique hobbies compared to other participants
S7	Tom Andelin (P07)	Concerts and Music, board games, family time	Tom is a bit older but very tech savvy.
S8	Maria (P08)	Cooking, Scooter-ing, being with friends	English is not her native language. Testing the accessibility of the language used.
S09	Paige Gordichuk (P09)	Music, trying new crafts, going out, problem solving	She likes to try new things to see if something enjoyable can fit into her busy schedule.
S10	Nami Robinson (P10)	Thrifting, Trying new food, fashion	Interested in trying new things with her boyfriend but has struggled thinking of them.
S11	Meesa Andelin (P11)	Paddleboarding, games, baking, crafts	Seemingly has a new favorite hobby every month and is a bit older (not as tech savvy)

Method

We ran a tree test with 10 tasks, each with a specific target page in the wireframe. 11 participants each did their own sort, excluding Ashley, Carley, and an AI agent. Each participant received this link, which leads to a page to enter their name and start the tree test. The tasks are shown to them one at a time with the opportunity for the participant to click through the page and complete the task. At every step there was the option to click “This is my answer”, indicating they think they’ve found the target page, or “I would give up”, which would send them to the next task. The results were logged to a google sheet, including the following fields: timestamp, participant, session start, task ID, task text, target page, first click label, first click destination, full click path, final destination, seconds elapsed, whether or not the participant gave up, and success. We looked through the raw data and noted trends and confusion points.

2. Tasks

#	Task	Target
1	Your roommate's been leaving a fresh sourdough loaf on the counter every Sunday. You want to try making your own starter.	Sourdough
2	You drove out to the desert last weekend and were blown away by the stars. You want to get into stargazing as a hobby.	Astronomy
3	A coworker keeps posting clips from the climbing gym, and you want to try it.	Rock Climbing
4	Your grandma made the quilt on your bed, and you want to learn to make something similar by hand.	Crochet, Embroidery, or Quilting
5	Finals week wrecked you, and a friend said she does a 10-minute meditation every morning to cope. You want to try something like it.	Meditation / Yoga
6	You've always wanted to build your own app on the side but never started.	Coding
7	Your friend group needs something fun for Friday night that isn't just watching a movie.	Escape Rooms
8	Your brother printed a custom phone stand at school, and you want to design and print your own stuff at home.	3D Printing
9	You want something active you can do with a couple of	Disc Golf or

	friends at the park on a Saturday, nothing that needs a full team.	Pickleball
10	You've been watching videos of people finding cool vintage furniture at Deseret Industries and flipping it. You want to get into that.	Thrifting/Antiquing

Each task is written as a short, everyday scenario that contains a reason for wanting the information, rather than a direct instruction. The tasks cover a range of hobbies in order to test category coverage.

3. Results

#	Task	Target	Most Common 1st Click	Most Common Final Dest.	Success Rate	Give-up Rate	Avg Time
1	Your roommate's been leaving a fresh sourdough loaf on the counter every Sunday. You want to try making your own starter.	Sourdough	Learning (7/11)	Sourdough (4)	5/11	3	41s
2	You drove out to the desert last weekend and were blown away by the stars. You want to get into stargazing as a hobby.	Astronomy	Outdoors (8/11)	Astronomy (4)	9/11	0	24s
3	A coworker keeps posting clips from the climbing gym, and you want to try it.	Rock Climbing	Movement (8/11)	Rock climbing (7)	9/11	0	13s
4	Your grandma made the quilt on your bed, and you want to learn to make something similar by hand.	Crochet, Embroidery, or Quilting	Creative Outlets (6/11)	Quilting (6)	7/11	0	21s
5	Finals week wrecked you, and a friend said she does a 10-minute meditation every morning to cope. You want to try something like it.	Meditation / Yoga	Movement (4/11)	Meditation (6)	10/11	0	21s
6	You've always wanted to build your own app on the side but never started.	Coding	Learning and Creative Outlets (both 5/11)	Coding (6)	9/11	0	19s

7	Your friend group needs something fun for Friday night that isn't just watching a movie.	Escape Rooms	Games (9/11)	The games main page (6)	8/11	0	12s
8	Your brother printed a custom phone stand at school, and you want to design and print your own stuff at home.	3D Printing	Creative Outlets (6/11)	3d printing (6)	7/11	1	29s
9	You want something active you can do with a couple of friends at the park on a Saturday, nothing that needs a full team.	Disc Golf or Pickleball	Outdoors (6/11)	The outdoors main page (4)	7/11	0	24s
10	You've been watching videos of people finding cool vintage furniture at Deseret Industries and flipping it. You want to get into that.	Thrifting/Antiquing	Creative Outlets (6/11)	thrifting/antiquing (4)	5/11	2	21s

Listening Notes

- One interesting thing we learned from one of our participants, who also completed a card sort, was that the way she grouped things together using cards, did not match the way she categorized hobbies in this test. She said “Once I found the tasks that I was looking for, I could tell that you grouped them together how I did with the cards, but now that I’m looking at them on a website it doesn’t make sense to me anymore.” This was especially true with the “Cultivating” category. It was the least clicked category in our tree test because it didn’t make sense anymore. When viewing the entire Cultivating group as a whole, the label makes sense. But it doesn’t draw users in on the first click.
- Some consistent comments made throughout the click tests were statements like, “oh I’m dumb” or “I should have known to look there”. We’d reassure them that they can’t do anything wrong and the goal is to learn from them. That tells us that even though our groupings make logical sense, the vibes and the labels are off. Especially for the “learning”, “cultivating”, and “media” labels.
- In Isaiah’s tree test study he made an interesting comment. He was completing the 3 tasks to look for rock climbing and had navigated to the “movement” section which was correct, but while staring at the page he said “I feel like it should be here” and then clicked away from the movement page. Rock climbing was on that page; he just didn’t see it! This told us we need more facets on this page to break up the lists of hobbies there. This could also be said for the outdoors section, which has a similar structure.

4. Analysis

Task 1: Sourdough

The “learning” label stole a lot of the first clicks that were meant for “cultivating” in this task. 7 out of 11 participants clicked on learning despite that being the wrong category. This told us that a lot of our participants view sourdough bread making as a skill to learn instead of a part of “homestead”-ing. The “cultivating” label isn’t signaling food-making hobbies to people.

Task 4: Quilting

Quilting had a similar dynamic to the sourdough task because the “learning” tab pulled 4 out of the 11 first clicks. Quilting requires a lot of skill and registers to users as something they want to learn how to do instead of an artsy craft to do. This misunderstanding between the learning and creative outlets tab shows that the categories are too vague. We should refine those labels and the facets if necessary too.

Task 8: 3D Printing

Just like the last two tasks, the relationship between the “learning” and “creative outlets” categories confused the participants. “Creative outlets” pulled away 6 out of the 11 first clicks which is substantial data because that is the majority. This shows us that we need to refine these labels because it’s a blaring issue at this point! 3D printing is actually supposed to be in learning. Out of the 6 participants that entered creative outlets, none of them ever clicked over to the learning section to even see what was in there. In this case 3D printing may need to be moved to the creative outlets area because even though it seems technical, the participants view it as creative.

Task 10: Thrifting/Antiquing

For this task, 6 out of 11 of our participants clicked into “creative outlets” but then 2 of those people went into the art facet instead of collecting. The “collecting” label was underneath the art label so it's possible that users clicked the first button that reminded them of thrifting. Or, perhaps collecting isn’t the correct place to put this hobby. We may need to update the collecting label to be more inclusive to second hand shopping.

Tasks where participants picked very different paths:

- **Task 2: Astronomy -** This task had 8/11 click to outdoors. We definitely need to move it to outdoors. The Other 3 users all went to different sections: learning, cultivating, and games. Despite all the different paths, this task had a high

success rate (82%) meaning that most participants did find astronomy in learning.

- **Task 5: Meditation -** Meditation was split in three different ways: movement pulled 4 clicks, cultivating pulled 3, and learning pulled 3 clicks. This split shows some ambiguity about where this should belong.
- **Task 6: Coding -** Coding was interesting because it had a 50-50 split between the learning and creative outlets pages. This was the only true tie in our data! Coding does belong with technology but this shows that the labels don't represent their contents as well as they should, especially "learning" since that's where coding belongs.

5. Design Decisions

Decision	Evidence	Why
Keep Gaming Label	9/11	9/11 Participants first clicked "Games" correctly
Rename Cultivating to Home and Garden	7/11	Participants clicked "learning" instead of "cultivating". The label didn't communicate what was inside of it
Add new facets under home and garden (home craft, kitchen and food, plants and growing, backyard and keeping)	5/11	That evidence shows the lowest success rate in the study for reaching the correct category. Homestead didn't draw the attention it needed to.
Reorder the facets inside of Creative Outlets: put collecting above art	3/6	Participants that clicked on creative outlets clicked art (3/6) instead of collecting. Collecting being put first could help draw more traffic to it since art seems like a catch all right now.
Rename "learning" to Education and Tech	5/11	That stat shows the split between users that click learning and creative outlets for hobbies inside of learning. That split is too even and the label needs to be specified.
Add facets under outdoors. (On the water, On land and trails, Air and thrill)	6/11	6 out of 11 participants clicked on movement after not being able to find their task under outdoors. The information needs additional structure to reduce cognitive load on the user, that way they'll be able to find their hobby easier.
Move astronomy to outdoors and put under	8/11	That many participants clicked on outdoors to find astronomy first.

air and thrill		
Add facets under movement. (Wellness, sports and competition, fitness and training, wheels and tricks)	Task five split in 3 ways! With only 4/11 in movement	Adding these facets under movement helps resolve some of the ambiguity under the category. There is a lot of content under this page and including these facets will draw focus to the areas we want to.
Move “meditation” from cultivating to movement	4/11	4/11 was the stat for the most clicked first click for meditation and that first click was movement.
Split media label into music and performance and film and audio	No specific stat here	We inferred the need for this split because media was vague and needed to be renamed. Instead of renaming with another vague label we decided to split this category into 2 more specific categories.
Add facets to “art” (visual art, words and story, novelty, craft, fashion and beauty)	Art had 26 hobbies inside.	26 items inside of this facet forced the participants to search for what they were looking for. The site is supposed to be suggestive, not exhausting.
Add facets under “craft” (needle and thread, paper, make and build)	No specific stat here	After adding facets to craft felt like the right thing to do after revamping our facets for other similar categories with lots of content.

Indented Text Outline of New Hierarchy:

Outdoors

- **NEW**: On the Water (Fishing, Jet Skiing, Hydrofoils, Boating, Paddleboarding, Sailing, Snorkeling, Freediving, Whitewater Rafting)
- **NEW**: On Land & Trails (Camping, RVing, Horseback Riding, Snowshoeing, Hiking/Trail Walking, Fossil Hunting, Birdwatching)
- **NEW**: Air & Thrill (Skydiving and Astronomy)
    - Astronomy (**MOVED** from Education & Tech)

Movement

- **NEW:** Wellness & Mind-Body (Pilates, Yoga, Calisthenics)
- Meditation (**MOVED** from Home & Garden)
- **NEW**: Sport & Competition (Pickleball, Bowling, Disc Golf, Martial Arts, Fencing, Table Tennis, Rock Climbing)

- **NEW**: Fitness & Training (Swimming, Cycling, Running, Weightlifting)
- **NEW**: Wheels & Tricks (Skateboarding, BMX, Parkour, Trampolining)

Home & Garden (**RENAMED** from Cultivating)

- **NEW**: Plants & Growing (Gardening, Plant Propagation, Bonsai)
- **NEW**: Kitchen & Food (Sourdough, Canning, Cake Decorating, Barbecue/Smoking)
- **NEW**: Home Craft (Candle Making, Soap Making)
- **NEW**: Backyard & Keeping (Chicken Keeping, Beekeeping)
- Homestead (**REMOVED**, contents redistributed into above facets)
- Meditation (**REMOVED**, **MOVED** to Movement and Wellness & Mind-Body)

Creative Outlets

- Collecting (**REORDERED** above Art)
- Bag Charms (**MOVED** from Art)
- Music & Performance (**RENAMED** from Media) (Guitar, Voice Lessons, Beatboxing, DJing, Concerts)
- **NEW**: Film & Audio, split from Media (Photography, Videography, Podcasting)
- **NEW**: Visual Art (Painting, Graphic Design, Coloring)
- **NEW:** Words & Story (Poetry, Journaling)
- **NEW**: Novelty (Juggling, Balloon Animals)
- **NEW:** Craft
    - **NEW:** Needle & Thread (Quilting, Crochet, Embroidery)
    - **NEW:** Paper & Flat (Scrapbooking, Calligraphy, Flower Pressing)
    - **NEW:** Build and Create (Pottery, Air Dry Clay, Resin Art, Origami, Model Kit Building, Slime Making, Bedazzling, Tie-Dye)
- **NEW:** Fashion and Beauty (Cosplaying, Nail Art, Makeup Artistry, Jewelry Making)

Education & Tech (**RENAMED** from Learning)

- Astronomy (**REMOVED** — **MOVED** to Outdoors and Air & Thrill)

Games *(unchanged)*

6. What you are choosing to ignore.

Appendix (*Full task-by-task results table, click path log, participant list)*

Table A1

Participants

Sort	Participant
S1	Kate Thornton (P01)
S2	Spencer Tagge (P02)
S3	Isaiah Pack (P03)
S4	Kyle Andelin (P04)
S5	Savannah Sloan (P05)
S6	Sarah G (P06)
S7	Tom Andelin (P07)
S8	Maria (P08)
S09	Paige Gordichuk (P09)
S10	Nami Robinson (P10)
S11	Meesa Andelin (P11)

Note. This table shows the participants.

Table A2

Tree Test Trends

#	Task	Target	Most Common 1st Click	Most Common Final Dest.	Success Rate	Give-up Rate	Avg Time
1	Your roommate's been leaving a fresh sourdough loaf on the counter every Sunday. You want to try making your own starter.	Sourdough	Learning (7/11)	Sourdough (4)	5/11	3	41s
2	You drove out to the desert last weekend and were blown away by the stars. You want to get into stargazing as a hobby.	Astronomy	Outdoors (8/11)	Astronomy (4)	9/11	0	24s
3	A coworker keeps posting clips from the climbing gym, and you want to try it.	Rock Climbing	Movement (8/11)	Rock climbing (7)	9/11	0	13s
4	Your grandma made the quilt on your bed, and you want to learn to make something similar by hand.	Crochet, Embroidery, or Quilting	Creative Outlets (6/11)	Quilting (6)	7/11	0	21s
5	Finals week wrecked you, and a friend said she does a 10-minute meditation every morning to cope. You want to try something like it.	Meditation / Yoga	Movement (4/11)	Meditation (6)	10/11	0	21s
6	You've always wanted to build your own app on the side but never started.	Coding	Learning and Creative Outlets (both 5/11)	Coding (6)	9/11	0	19s
7	Your friend group needs something fun for Friday night that isn't just watching a movie.	Escape Rooms	Games (9/11)	The games main page (6)	8/11	0	12s
8	Your brother printed a custom phone stand at school, and you want to design and print your own stuff at home.	3D Printing	Creative Outlets (6/11)	3d printing (6)	7/11	1	29s

9	You want something active you can do with a couple of friends at the park on a Saturday, nothing that needs a full team.	Disc Golf or Pickleball	Outdoors (6/11)	The outdoors main page (4)	7/11	0	24s
10	You've been watching videos of people finding cool vintage furniture at Deseret Industries and flipping it. You want to get into that.	Thrifting/Antiquing	Creative Outlets (6/11)	thrifting/antiquing (4)	5/11	2	21s

Note. This table shows common trends across participants in the tree test.

Table A3

Click Log and Raw Data

https://docs.google.com/spreadsheets/d/1qbGFXHJojl10tiGsD47zOs1FHLgFHI1Y9epQ5iWOGDQ/edit?usp=sharing

Note. This link redirects you to a google sheet with our raw data collected in each tree test.